Supporting Information:

Beyond chronological age: Modeling human growth trajectories using dental development stages

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1 Model Description

This section of the Supporting Information describes the Bayesian growth models. Across all models, y is a scalar response variable and x is some predictor that can take on a scalar, integer, or ordinal value. Below, I consider individual growth models that fix the response variable at some growth trait (e.g., maximum diaphyseal length of the femur) but modify the characterization of the predictor x depending on model type. All data stems from (Stull and Corron 2022).

1.1 Motivation

While all code and analyses relate to the use of a single developmental variable (mandibular M1), the application of the specific monotonic effects model is wide ranging and may stem to not only other maturational and/or developmental markers, but could be expanded to scenarios outside of growth and development where monotonicity of the predictor is a presumed constraint – e.g., using taphonomic scores to predict a scalar accumulated degree days (Megyesi et al. 2005).

1.2 Chronological Age Model

1.2.1 Data Model

Let x be a scalar independent variable and let y be a scalar response variable such that

$$y \sim \mathcal{N}\big(\mu(x, \boldsymbol{\theta_\mu}), \sigma(x, \boldsymbol{\theta_\sigma})\big),$$

where $\mathcal{N}$ denotes the normal distribution, $\mu(x, \boldsymbol{\theta_\mu})$ is the mean, $\sigma(x, \boldsymbol{\theta_\sigma})$ is the standard deviation, $\boldsymbol{\theta_\mu}$ is a vector parameterizing the mean, and $\boldsymbol{\theta_\sigma}$ is a vector parameterizing the standard deviation.

1.2.2 Process Model

The mean μ is a function of chronological age x and vector of parameters $\boldsymbol{\theta_\mu} = [a \;\; r \;\; b]^T$ such that the growth curve is defined as an offset power law (Wolfe and Stull 2026; Stull et al. 2023).

$$\mu = a x^r + b$$

The standard deviation σ is a function of chronological age x and a linear combination of parameters $\boldsymbol{\theta_\sigma} = [\beta_0 \;\; \beta_1]^T$ such that

$\log(\sigma) = \beta_0 + \beta_1 x$

Note, this explicit specification assumes the residual standard deviation changes with respect to age – presuming heteroscedasticity in the error term. In brms, noise terms are sampled according to a logarithmic link in order to ensure positivity and identifiability.

1.2.3 Prior Model

Priors in the chronological age model are relatively informative and based on previous testing in Wolfe and Stull (2026).

$$a \sim normal(65,1)$$

$$r \sim normal(0,1)$$

$$b \sim normal(65,1)$$

$$\beta_0 \sim student(3,0,2.5)$$

$$\beta_1 \propto 1$$

Note, β_1 is an improper / flat prior and presumes support across all real numbers. Assuming reasonable sample sizes this may be appropriate in theory, but it is recommended to adjust based on values on dataset for future inference.

1.3 Monotonic Effects Model

1.3.1 Data Model

Let x be a positively monotonically constrained ordinal variable and let y be a scalar response variable such that

$$y \sim \mathcal{N}\left(\mu\left(x, \boldsymbol{\theta_\mu}\right), \sigma(x, \boldsymbol{\theta_\sigma})\right),$$

where $\mathcal{N}$ denotes the normal distribution, $\mu\left(x, \boldsymbol{\theta_\mu}\right)$ is the mean, $\sigma(x, \boldsymbol{\theta_\sigma})$ is the standard deviation, $\boldsymbol{\theta_\mu}$ is a vector parameterizing the mean, and $\boldsymbol{\theta_\sigma}$ is a vector parameterizing the standard deviation.

1.3.2 Process Model

I define K ordinal categories with D steps in between categories – where $D + 1 = K$. For any value $x \in \{0, \dots, D\}, x$ can be defined as $bmo(\zeta, k_i)$ or a monotonic transform. The mean function can be further defined as

$$\mu = \beta_0^{\mu} + b^{\mu} D \sum_{k=1}^{K} \zeta_k^{\mu},$$

where β_0 is the expected response at the baseline category, k is each individual category, and ζ_k is a simplex parameter at each step between the categories. The sum of all simplex parameters $\sum_{i=1}^{D} \zeta_i = 1$. The total expected change from lowest to highest category is bD where b is the expected change in between categories. The standard deviation is parameterized in much the same way on the log scale ensuring positivity and heteroscedasticity with increasing stage

$$\log(\sigma) = \beta_0^{\sigma} + b^{\sigma} D \sum_{k=1}^{K} \zeta_k^{\sigma}.$$

1.3.3 Prior Model

All priors were kept default from initial brms settings including the improper flat prior on the slope terms.

$$\beta_0^{\mu} \sim student(3, 159.7, 93.2)$$

$$\beta_0^{\sigma} \sim student(3, 0, 2.5)$$

$$\zeta^{\mu} \sim Dirichlet(1)$$

$$\zeta^{\sigma} \sim Dirichlet(1)$$

$$b^{\mu} \propto 1$$

$$b^{\sigma} \propto 1$$

The Dirichlet concentration vector assumes a value of 1 across all steps [12 total in dental development scores]. This equates to a uniform distribution over the simplex.

1.4 Integer Predictor Model

1.4.1 Data Model

Let x be a continuous numeric [integer] variable and let y be a scalar response variable such that

$$y \sim \mathcal{N}\left(\mu(x, \boldsymbol{\theta}_{\boldsymbol{\mu}}), \sigma(x, \boldsymbol{\theta}_{\boldsymbol{\sigma}})\right),$$

where $\mathcal{N}$ denotes the normal distribution, $\mu(x, \boldsymbol{\theta}_{\boldsymbol{\mu}})$ is the mean, $\sigma(x, \boldsymbol{\theta}_{\boldsymbol{\sigma}})$ is the standard deviation, $\boldsymbol{\theta}_{\boldsymbol{\mu}}$ is a vector parameterizing the mean, and $\boldsymbol{\theta}_{\boldsymbol{\sigma}}$ is a vector parameterizing the standard deviation.

1.4.2 Process Model

The integer model is a Gaussian location-scale model with linear predictors on both the mean and standard shifts the mean and standard deviation by a constant amount. The mean can be written as

$\mu = \beta_0^\mu + x^T \beta_1^\mu,$

where β_0^μ is an intercept term and $x^T \beta_1^\mu$ is a linear predictor describing the effect of developmental stage on femoral growth. Similarly, the standard deviation can be written as

$\sigma = \beta_0^\sigma + x^T \beta_1^\sigma,$

where β_0^σ is an intercept term describing baseline residual standard deviation and $x^T \beta_1^\sigma$ is a linear predictor describing the change in residual standard deviation with increasing stage.

1.4.3 Prior Model

All predictors are centered and scaled prior to fit (default in *brms;* Bürkner 2017) and the priors reflect this convention. Again, all priors were kept default from initial brms settings including the improper flat prior on the slope terms.

$$\beta_0^\mu \sim student(3, 159.7, 93.2)$$

$$\beta_0^\sigma \sim student(3,0,2.5)$$

$$\beta_1^\mu \propto 1$$

$$\beta_1^\sigma \propto 1$$

1.5 Factor Predictor Models – Unordered and Ordered

1.5.1 Data Model

$$y \sim \mathcal{N}\big(\mu(x, \boldsymbol{\theta_\mu}), \sigma(x, \boldsymbol{\theta_\sigma})\big)$$

1.5.2 Process Model

$$\mu = \beta_0^\mu + x^T \beta_1^\mu$$

$$\sigma = \beta_0^\sigma + x^T \beta_1^\sigma$$

1.5.3 Prior Model

$$\beta_0^\mu \sim student(3, 159.7, 93.2)$$

$$\beta_0^\sigma \sim student(3,0,2.5)$$

$$\beta_1^\mu \propto 1$$

$$\beta_1^\sigma \propto 1$$

1.5.4 Explanation

The factor-coded models are identical in construction to the integer model described above – factors are imported into the model as integer values. As a point of fact, the exact model text as translated from *brms* is identical when importing ordered or unordered values. The distinction between models lies in the interpretation of the predictor values. In the unordered model, developmental stage is dummy coded 0/1 and modeled using treatment contrasts where coefficients represent contrasts of each stage relative to the reference stage. In the ordered model, developmental stage was modeled as an ordered factor using orthogonal polynomial contrasts representing polynomial trends between equally spaced levels. This contrast could go up or done – therefore, not strictly monotonic in nature.

2.0 Model fitting and diagnostics

All models were fitted in R (version 4.5.2) using the *brms* package (version 2.23.0) (Bürkner 2017, 2018, 2021), which compiles models to Stan (Stan Development Team 2026) and samples via cmdstanr (Gabry and Cesnovar 2022) based on cmdstan version 2.39.0. Posterior sampling used 4 Hamiltonian Monte Carlo chains with 2000 iterations each, the first 1000 discarded as warmup, yielding 4000 post-warmup draws per model.

Sampler control parameters were set to adapt_delta = 0.95 and max_treedepth = 10 to reduce divergent transitions in the non-linear and distributional models. Priors are as specified above. Predictors entering linear terms were mean-centered internally by brms, and reported intercepts were back-transformed to the original predictor scale.

Convergence was assessed by the potential scale reduction factor ($\hat{R} \leq 1.01$ for all parameters), bulk and tail effective sample sizes (both in the hundreds–thousands range), and visual inspection of trace and rank plots. There were no divergent transitions after warmup, and no transitions saturated the maximum tree depth. Model fit was further evaluated with posterior predictive checks, and predictive performance was compared across models using approximate leave-one-out cross-validation (LOO-CV via PSIS) (Vehtari et al. 2024, 2021, 2017; Gelman et al. 2014).

3.0 References

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